MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
1800 SECURITY BASE CAMP
FIRE WATER TANK FOUNDATION

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TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1. INTENT 4

2. EXECUTIVE SUMMARY 4

3. SCOPE 4

4. DESIGN REQUIREMENTS 5

    4.1 Design Loads 5

5. CONCRETE FOUNDATIONS 5

    5.1 Design 5

    5.2 Materials 5

    5.3 Foundations for Aboveground Tanks 5

    5.4 Minimum Concrete Cover 5

    5.5 Allowable Soil Bearing Capacity 6

6. DESIGN CALCULATION FOR FOUNDATION 6

Reference 10

ATTACHMENT 1 LOAD DATA DRAWINGS 10

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to Prepare for design of **Fire Water Tank (1800)-** Tank's Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Load Data Provided.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Foundation and Load Plan (Doc. No. MD-216-39LA-EG-FG-GAD-1003) attached in Attachment 1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5 CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Aboveground Tanks

Foundation shall be Earth Foundation with a Concrete Ring wall. The design calculations are prepared as per recommendation of standard API 650 - Annex B.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	50

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5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 375 kN/m^2 at 1.0m depth below EGL (ie. RL 923.2) refer Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

6. DESIGN CALCULATION FOR FOUNDATION

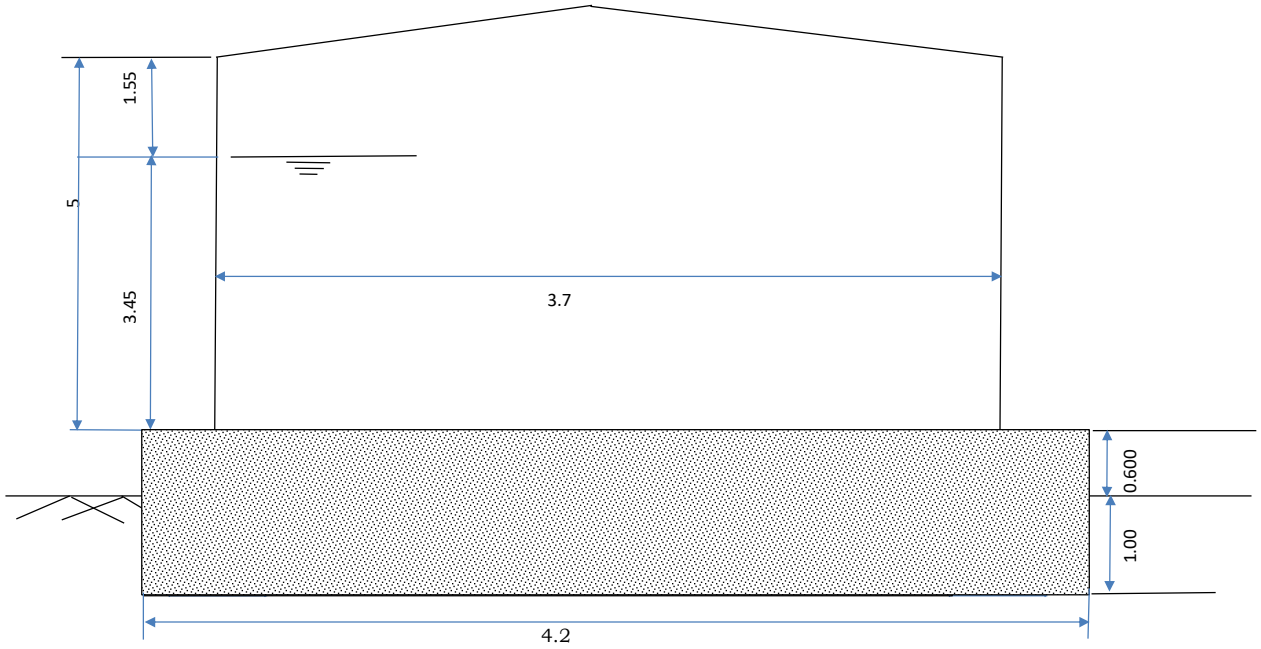
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Tank Loading Data

	Loading Type		
Weight, V	Empty (W)	96	kN
	Operating (Wo)	466	kN
	Test (Wt)	527	kN

Summary of Tank Specification



Diameter of Tank, D		=	3.7 m	
Height of Tank, H		=	5 m	
Height of Tank Fluid, H_L		=	3.45 m	
			3.2	
Outer Diameter of Foundation, OD		=	4.2 m	
Height of Concrete Foundation, h		=	1.60 m	
		=		
	B.C.D	=	3.95 m	
Top of Concrete		=	0.6 m	(Above Finished Level)
Unit Weight of Liquid, γ_L		=	9.95 kN/m^3	
Unit Weight of Concrete, γ_c		=	24 kN/m^3	

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Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	9.00	23.00
	Operating	9.00	23.00
Seismic Loads	Operating	8.00	17.00

Foundation Weight

a) Foundation Weight = 532.01 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 5.81 m^2

Full cross-sectional area for operating condition, Ao= 13.85 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 4.82 m^3

Sectional modulus for operating condition,Zo = 7.27 m^3

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301,2007 , Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	466.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	57.60	9.00	23.00
	Operating	LC103	D + F1 + W	466.00	9.00	23.00
		LC104	D + F1+ 0.7 E	466.00	5.60	11.90
		LC105	0.6 (D + F1) + 0.7 E	279.60	5.60	11.90
	Test	LC106	D + F2	527.00	0.00	0.00

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 375 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 498.75 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Qmax= V/A - M/Z kN/m^2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	1306.84	0.00	94.33	94.33	3.98	OK
LC102	589.61	37.40	109.20	93.69	4.57	OK
LC103	1306.84	37.40	99.47	89.18	5.01	OK
LC104	1306.84	20.86	97.19	91.46	5.13	OK
LC105	1120.44	20.86	83.74	78.00	5.96	OK
LC106	1367.84	0.00	98.73	98.73	3.80	OK

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Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear
Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	1306.84	0.4	522.74	0.00	-	-
LC102	589.61	0.4	235.84	9.00	26.2	OK
LC103	1306.84	0.4	522.74	9.00	58.1	OK
LC104	1306.84	0.4	522.74	5.60	93.3	OK
LC105	1120.44	0.4	448.18	5.60	80.0	OK
LC106	1367.84	0.4	547.14	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment
The resultant Safety Factor should be 1.5 minimum
Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	1306.84	2.1	2744.37	0.00	-	-
LC102	589.61	2.1	1238.18	37.40	33.1	OK
LC103	1306.84	2.1	2744.37	37.40	73.4	OK
LC104	1306.84	2.1	2744.37	20.86	131.6	OK
LC105	1120.44	2.1	2352.93	20.86	112.8	OK
LC106	1367.84	2.1	2872.47	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304,2018 , Table 5.3.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	652.40	0.00	10.75	4.97
Short term	Empty	LC102	0.9 D + 1.6 W	86.40	36.80	10.75	4.97
	Operating	LC103	1.2 (D + F1) + 1.6 W	559.20	36.80	10.75	4.97
		LC104	1.2 (D + F1) + E	559.20	17.00	10.75	4.97
		LC105	0.9 (D + F1) + E	419.40	17.00	10.75	4.97
	Test	LC106	1.4 (D + F2)	737.80	0.00	10.75	4.97

I) Horizontal Reinforcement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)
Fy = 420 MPa
Hence, Provide 12 Nos. D16= 2177.501 mm^2 OK

II) Vertical Reinforcement

Minimum As = 0.0015 * b * D = 750 mm^2 (As per API 650 , Cl. No. B.4.2.3)
So, Used As = 2 row * 5 ea - D12 (@ 200mm)
=1131mm2 > Required As OK

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REFERENCES

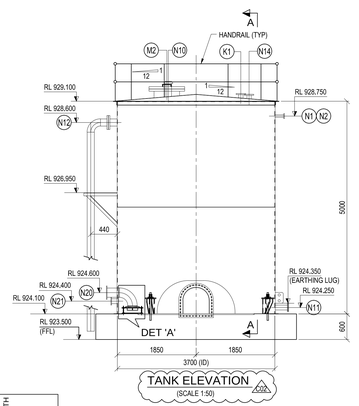
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

ATTACHMENT 1 LOAD DATA DRAWING

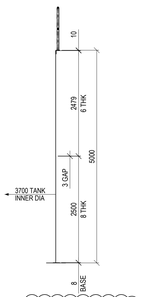
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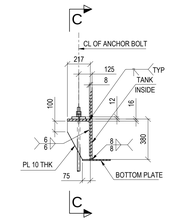
DRAWING NUMBER: MD-216-39LA-EG-FG-GAD-1003 REV D01



TANK ELEVATION
(SCALE 1:50) C02



SECTION A - A
(SCALE - 1: 50) C02



TYP DETAIL OF ANCHOR CHAIR
SCALE- 1:20 C02

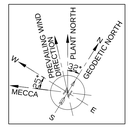
MATERIAL SPECIFICATION

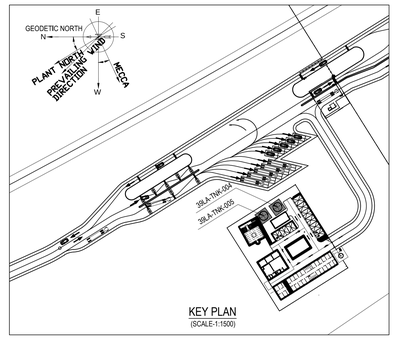
SHELL, BOTTOM, ANNULAR, ROOF	EN 10025 GR.S275 J0
COD, MANHOLE	EN 10025 GR.S275 J0
MANWAY NECKS/ MANWAY FLANGES	EN 10025 GR.S275 J0
NOZZLE NECKS (PIPES) (600 NB & BELOW)	ASTM A106 Gr.B
DELETED	DELETED C02
NOZZLE FLANGES/ BLIND FLANGE	SA 105
GASKETS FOR NOZZLES	EPDM C02
GASKETS FOR MANWAY/ COD	3mm THICK COMPRESSED NON-ASBESTOS SHEETS.
BOLTS/ NUTS FOR NOZZLE	SA 193 Gr. B7 / 194 Gr. 2H
BOLTS/ NUTS FOR MANWAY	SA 193 Gr. B7 / 194 Gr. 2H
STRUCTURAL BOLTS	ASTM A325M
ANCHOR CHAIR	EN 10025 GR.S275 J0
STRUCTURAL(ROLLED SECTION)-STAIRWAY / PLATFORM / ROOF SUPPORTING STRUCTURE	EN 10025 GR.S275 JR
HANDRAIL PIPE / HANDRAIL POST	ASTM A53 TYPE E, GRADE B
PIPE FITTINGS	SA 234 Gr. WPB
REINFORCEMENT PAD/GUSSET/ EXTERNAL SUPPORT ETC.	EN 10025 GR.S275 J0
EARTHING LUGS	EN 10025 GR.S275 J0
ANCHOR BOLTS	ASTM F1554 GR.36
CURB ANGLE	EN 10025 GR.S275 JR
NAME PLATE	STAINLESS STEEL(SS 316)

FOUNDATION LOAD DATA: ** C02

BASIC LOAD CASE	VERTICAL COMPRESSION (KN)	VERTICAL UPLIFT (KN)	HORIZONTAL SHEAR (KN)	OVERTURNING MOMENT (ABOUT SHELL BOTTOM JOINT) (KN-M)	SLAB MOMENT (KN-M)
DEAD LOAD (NOMINAL THICKNESS)	96	–	–	–	–
DEAD LOAD (CORRODED THICKNESS)	81	–	–	–	–
ROOF PIPE LIQUID LOAD	11	–	–	–	–
SHELL PIPE LIQUID LOAD	11	–	–	–	–
LIVE LOAD	26	–	–	–	–
TANK LIQUID LOAD (OPERATING)	466	–	–	–	–
TANK LIQUID LOAD (TEST)	527	–	–	–	–
WIND LOAD ON SHELL	–	–	9	23	–
WIND LOAD ON ROOF	–	9	–	17	–
SEISMIC LOAD	18	18	8	17	21

** REFER NOTE 15





KEY PLAN
(SCALE-1:1500)

NOZZLE SCHEDULE (FOR TANK NO 39LA-TNK-004) C02

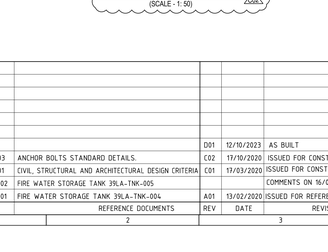
NOZZLE	SERVICE	SIZE (DN)	THICKNESS (SCHEDULE)	RATING #	TYPE	ELEVATION "H"	PROJECTION "P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA	RF PAD THK.	REMARK
N1	INLET-FILLING BY TANKER	100	Sch XS	CL 150	RFSO	4.650	200	-	135°	SHELL	305	8	
N2	RECIRCULATION LINE	100	Sch XS	CL 150	RFSO	4.650	200	-	225°	SHELL	305	8	
N10	TANK VENT	150	-	COVER ARRANGEMENT		-	300	750	240°	ROOF	400	8	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.150	300	-	285°	SHELL	305	8	
N12	OVERFLOW	150	Sch XS	CL 150	RFSO	4.500	200	-	285°	SHELL	400	8	
N13	CLEAN OUT DOOR (COD)-hxb	600x600	-	API 650		-	-	-	315°	SHELL	915X2035	8	
N14	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	100	1375	65°	ROOF	305	8	
N20	FIRE WATER OUTLET	200	Sch XS	CL 150	RFSO	0.500	250	-	270°	SHELL	485	8	
N21	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	0.300	200	-	240°	SHELL	305	8	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	100	1375	50°	ROOF	305	8	
M1	MANHOLE C/W DAVIT	750	-	API 650		0.925	175	-	90°	SHELL	1545	8	
M2	MANHOLE C/W HINGED COVER	600	-			-	300	1375	330°	ROOF	1150	8	

NOZZLE SCHEDULE (FOR TANK NO 39LA-TNK-005) C02

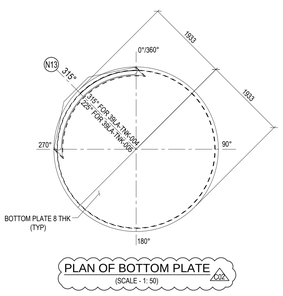
NOZZLE	SERVICE	SIZE (DN)	THICKNESS (SCHEDULE)	RATING #	TYPE	ELEVATION "H"	PROJECTION "P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA	RF PAD THK.	REMARK
N1	INLET-FILLING BY TANKER	100	Sch XS	CL 150	RFSO	4.650	200	-	135°	SHELL	305	8	
N2	RECIRCULATION LINE	100	Sch XS	CL 150	RFSO	4.650	200	-	315°	SHELL	305	8	
N10	TANK VENT	150	-	COVER ARRANGEMENT		-	300	750	240°	ROOF	400	8	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.150	300	-	150°	SHELL	305	8	
N12	OVERFLOW	150	Sch XS	CL 150	RFSO	4.500	200	-	150°	SHELL	400	8	
N13	CLEAN OUT DOOR (COD)-hxb	600x600	-	API 650		-	-	-	225°	SHELL	915X2090	8	
N14	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	100	1375	65°	ROOF	305	8	
N20	FIRE WATER OUTLET	200	Sch XS	CL 150	RFSO	0.500	250	-	270°	SHELL	485	8	
N21	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	0.300	200	-	225°	SHELL	305	8	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	100	1375	50°	ROOF	305	8	
M1	MANHOLE C/W DAVIT	750	-	API 650		0.925	175	-	0°	SHELL	1545	8	
M2	MANHOLE C/W HINGED COVER	600	-			-	300	1375	330°	ROOF	1150	8	

DESIGN DATA C02

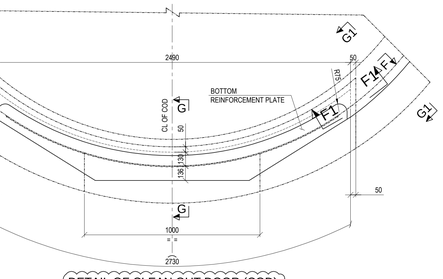
TANK DETAILS	LOCATION		SECURITY BASE CAMP
	TANK NAME		FIRE WATER TANK
	EQUIPMENT TAG No.		39LA-TNK-004 & 39LA-TNK-005
	TANK DIMENSIONS		INSIDE DIAMETER : 3700mm, HEIGHT: 5000mm
	QUANTITY (NUMBER)		2
	ROOF TYPE & SLOPE		SLOPED CONE ROOF (SLOPE 1 IN 12) CONICAL APEX UP
	BOTTOM TYPE & SLOPE		SLOPED BOTTOM (SLOPE 1 IN 120) CONICAL APEX UP
	FIELD ERECTED/SHOP FABRICATED		SHOP FABRICATED
	STORED FLUID		FIRE WATER
DESIGN CODE	DESIGN CODE		API 650-TWELFTH EDITION, MARCH 2013, ADDENDUM 3, AUGUST 2018
OPERATING DATA	OPERATING TEMPERATURE		40° C
	NET WORKING CAPACITY		33 m^3
	MAXIMUM CAPACITY		47 m^3
	OPERATING LEVEL	MINIMUM	150 mm
		MAXIMUM	4410 mm
	OPERATING DENSITY OF CONTENT		1.0 t/m^3
MECHANICAL & PROCESS DATA	DESIGN TEMPERATURE		50° C
	DESIGN PRESSURE (INTERNAL/EXTERNAL)		ATM/ATM
	TEST PRESSURE		ATM
	HYDRO STATIC TEST PRESSURE		FULL OF WATER
	DESIGN DENSITY OF CONTENT		1.0 t/m^3
	TESTING		AS PER API 650
	INSPECTION		AS PER API 650
	DENSITY OF SCALE		N/A
	DESIGN SCALE THICKNESS	TANK SHELL/ROOF	N/A
	DESIGN FILL RATE		150 m/hr (TBC)
	DESIGN EMPTY RATE		120 m/hr (TBC)
	CORROSION ALLOWANCE	SHELL	1.6 mm
		ROOF PLATE	0 mm
		BOTTOM PLATE / ANNULAR PLATE	1.6 mm
		ANCHOR BOLT	0 mm
		STRUCTURAL	1.6 mm
WIND LOAD DATA	BASIC WIND SPEED		43 m/sec. (155 Km/hr.)
	SITE EXPOSURE		C
	VELOCITY PRESSURE EXPOSURE COEFFICIENT		K_z = AS PER ELEVATION
	TOPOGRAPHIC FACTOR		K_{zt}=1.0
	WIND DIRECTIONALITY FACTOR		K_d =0.95
SEISMIC LOAD DATA	SITE CLASS		B
	SEISMIC USER GROUP		SUG - I
	IMPORTANCE FACTOR	(I)	1
	SCALING FACTOR	(Q)	2/3
	MAXIMUM DESIGN LIQUID LEVEL	(H)	4.420 m
	CO-EFFICIENT TO ADJUST THE SPECTRAL ACCELERATION FROM 5% TO 0.5% DAMPING	(K)	1.50
	SEISMIC CO-EFFICIENTS		S_{ds}= 0.1g, S_{d1}= 0.02g, F_a = 1.0, F_v = 1.0 S_s = 0.067g, S_1 = 0.013g, T = 4 sec.
OTHER DATA	ROOF LIVE LOAD		1.2 kPa
	FOUNDATION TYPE (RING WALL/SLAB)		RING WALL
	PWHT REQUIRED(YES/NO)- IF YES,SPECIFY REASON FOR PWHT		NO
	REASON FOR PWHT(PROCESS REASON/PER API CODE)		AS PER API 650
	RADIOGRAPHY		AS PER CL 8.1 - API 650
	AGITATOR RAKE REQUIRED	(YES/NO)	NO
		EQUIPMENT TYPE (AGITATOR/RAKE)	N/A
		AGITATOR/RAKE SUPPORTED BY TANK (YES/NO)	N/A
	INSULATION	SHELL	N/A
		ROOF	N/A
	LINER REQUIRED	(YES/NO)	NO
		LINER MATERIAL	N/A
		LINER THICKNESS (ROOF/SHELL/BOT)	N/A
	PAINTING	INTERNAL	YES
		EXTERNAL	YES (REFER NOTE 12)
	PIPE SUPPORT		YES
	CAGE LADDER		YES
	ROOF CIRCUMFERENTIAL HANDRAIL		YES
	WIND GIRDER		N/A
	CATHODIC PROTECTION		NO
	LEAK DETECTION (APPURTENANCES)		NO
	INTERNAL BAFFLES		NO
	EARTHING LUGS		YES



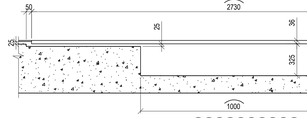
PLAN
(SCALE - 1: 50) C02



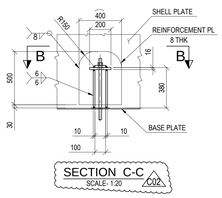
PLAN OF BOTTOM PLATE
(SCALE - 1: 50) C02



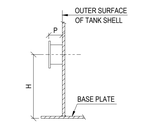
DETAIL OF CLEAN OUT DOOR (COD)
(SIZE 600x600)
SCALE- 1:20 C02



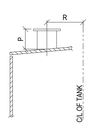
SECTION B-B
SCALE- 1:20 C02



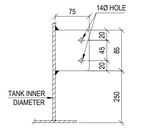
SECTION C-C
SCALE- 1:20 C02



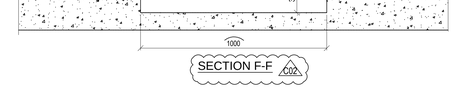
SHELL NOZZLES / FITTINGS
C02



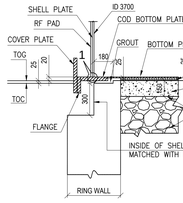
ROOF NOZZLES / FITTINGS
C02



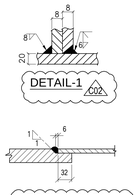
DETAIL OF EARTHING LUGS (2 NOS)
(QUANTITY : 2 Nos.) C02

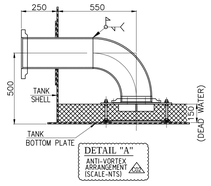


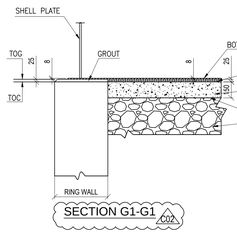
SECTION F-F
C02



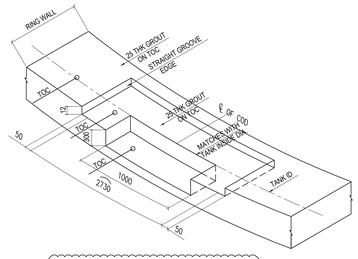
SECTION G-G
C02







SECTION G1-G1
C02



3D VIEW OF FOUNDATION NOTCH
(FOR COD) A

LEGEND:

ID - INNER DIAMETER
OD - OUTER DIAMETER
COD - CLEAN OUT DOOR
LG - LONG
FGL - FINISHED GROUND LEVEL
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
RL - REDUCED LEVEL
RF - REINFORCEMENT
TOC - TOP OF CONCRETE
GST - GUSSET
PLT - PLATE
STIFF - STIFENER
THK - THICKNESS
CLT - CLEAT
UNO - UNLESS NOTED OTHERWISE
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
NOZ - NOZZLE
DN - DIAMETER NOMINAL
SCH - SCHEDULE
ELVN - ELEVATION
PROJ - PROJECTION
MM - MILLIMETER
TYP - TYPICAL
GA - GENERAL ARRANGEMENT
DRG - DRAWING
CL - CENTER LINE
TOS - TOP OF STEEL
WP - WORKING POINT
N/S - NEAR SIDE
F/S - FAR SIDE
DIA - DIAMETER
C/F - CUT FORM
TBC - TO BE CONFIRMED
TOG - TOP OF GRATING
N/A - NOT APPLICABLE
ATM - ATMOSPHERIC
MIN - MINIMUM

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. BOLT HOLES IN NOZZLE FLANGES & MANHOLES SHALL STRADDLE VERTICAL CENTER LINES ON SHELL & NORTH SOUTH CENTERLINES ON ROOF.
5. NOZZLE PROJECTION ON SHELL SIDE REFERS FROM EXTERNAL SURFACE OF SHELL PLATE TO FACE OF FLANGE. NOZZLE PROJECTION ON ROOF SIDE REFERS FROM TOP OF ROOF PLATE TO FACE OF FLANGE.
6. VERTICAL SHELL JOINTS SHOULD PREFERABLY BE OFFSET BY 1000 MM.
7. CURB ANGLE JOINTS SHALL STAGGER ATLEAST 300 MM WITH VERTICAL SHELL PLATE WELD.
8. REINFORCEMENT PADS TO HAVE TELLTALE HOLES OF 6 MM DIA AND SHALL BE TESTED AT 100 kPa PRESSURE WITH SOAP SOLUTION AND THEN FILLED WITH HARD GREASE.
9. FLANGE (COD & MANHOLE) GASKET SEATING SURFACE SHALL BE PROVIDED WITH 0.0032 TO 0.0063 mm (3.2 TO 6.3 µm) ROUGHNESS FINISH.
10. COD GROOVE DIMENSIONS ARE FINALISED BASED ON COD BOTTOM PLATE DIMENSIONS.
11. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD-216-39LA-EG-FG-GAD-1004 & 1005.
12. PAINTING SHALL CONFORM TO MAADEN SPECIFICATION NUMBER MD-101-SMEM-EG-ME-SPC-0007
13. ALL NOZZLE FLANGES SHALL BE IN ACCORDANCE WITH ASME B16.5 FOR SIZE UPTO DN600 AND TO ASME B16.47 SERIES A FOR DN750 AND ABOVE.
14. FOR HANDRAIL REFER DRG. NO. MD-216-1000-EG-ST-DED-1002 & 1003.(LATEST). FOR CAGE LADDER REFER DRG. NO. MD-216-1000-EG-ST-DED-1005 & 1006 (LATEST).
15. ALL LOADS ARE "UNFACTORED". C02

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	12/10/2023	AS BUILT	SOH	PRI	KDU	AG	
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	C02	17/10/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C02	SOH	PRI	KDU	AG	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	C01	17/03/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER'S	SJH	PRI	KDU	AG	
MD-216-39LA-EG-FG-GAD-1002	FIRE WATER STORAGE TANK 39LA-TNK-005			COMMENTS ON 16/03/2020 & REVISED AS MKD.					
MD-216-39LA-EG-FG-GAD-1001	FIRE WATER STORAGE TANK 39LA-TNK-004	A01	13/02/2020	ISSUED FOR REFERENCE	SJH	PRI	KDU	AG	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER	019045_080_S_G_26130_001		REVISION 3
OUTOTEC DRAWING NUMBER	900707__GAU01__AE01__MTL01__39103	DOCUMENT ID	OU100673480 REVISION 3
DRAWN	SJH	10/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	PRI	12/02/2020	GENERAL ARRANGEMENT
ENGINEER	KDU	11/02/2020	FIRE WATER STORAGE & DISTRIBUTION
APPROVED	AG	13/02/2020	PLAN, ELVN, DETAIL OF TANK 39LA-TNK-004&005
EMPLOYER			
SCALE	1:100,20	A0	DRAWING NUMBER: MD - 216 - 39LA - EG - FG - GAD - 1003 REV: D01